

Managing Disruptive Technologies – Assignment 5

Please mention your name, Andrew id and section. The assignment is to be done individually. It is due by 11:59pm on the 28th of April. Submission through Gradescope.

Objective: The main goal of the assignment is to dig deeper into the concepts of data network effects and returns to data for ML models that have been introduced in Lectures 10 and 11. Moreover, to bring about practical relevance, we will analyze real data output provided to make business decisions about the use of algorithms/ML models within a company. Eventually, after analyzing the data, you will decide about whether and when to use algorithmic recommendations.

QUESTION 1. Your answer to this question should be **at most 4 pages** long (that is, 2 two-sided pages) including figures, tables and references as necessary. APA referencing style is preferred. We will only give you credit for answers supported by evidence. If you add computations, tables or figures without comments or discussion, you will not receive credit for any insights one might glean from them. Likewise, you do not give credit for unsubstantiated claims (e.g. without computations, tables or figures) in the text file. Font size 11 and 1.5 spacing. **(100 points)**

In Lecture 10, we carried out a data exercise where we analyzed the impact of algorithmic recommendations relative to human creation for an online media platform (“NEWS”). **For details about the context, refer to the Background document for Lecture 10.** In this exercise, I want to continue on that example and analyze some output from the full sample of more than 130 million observations. This exercise is similar in part to what we did in the lecture. As a reminder, the A/B test had one treatment and one control condition. The control condition had a human curating the online page which would be the same for anyone who arrived at any particular point in time. In the treatment group, a machine learning model (algorithm) personalizes recommendation on one slot on the page (slot 4). The ML model is a hybrid one which uses a combination of information filtering and collaborative filtering to recommend content. Collaborative filtering, based on similar people on the platform, is used if the particular user has not been seen before. If a user visits the webpage several times, the algorithm utilizes data from

past visits to tweak its recommendations. The variable Treatment=1 if the user is in the treatment condition and =0 if a user is in the control condition. The outcome variable of interest is the number of clicks on the treated slot in a user-session (slot 4). The output from the full A/B test is as follows:

Table 1: Performance of Algorithmic Recommendations Relative to Human Curation

	Outcome: Clicks on Slot 4		
	(1)	(2)	(3)
Treatment	0.0007*** (0.00004)	-0.0081*** (0.00008)	-0.0125*** (0.00014)
Treatment $\times$ Log(Prior Visits)		0.0030*** (0.00003)	0.0084*** (0.00013)
Treatment $\times$ Log(Prior Visits) ²			-0.0009*** (0.00002)
Constant	0.0277	0.0277	0.0277
Observations	137,689,847	137,689,847	137,689,847

Note: The dependent variable is the number of clicks on slot 4. The unit of observation is user-session. *Log Prior Visits* is the natural logarithm of user *i*'s prior visits to the homepage at time *t*. Standard errors in parentheses. *** means p-value < 0.001

As the CTO of the company, you have to execute decisions based on the results of this experiment. We will try to think of the best strategy as we move sequentially from columns (1) to (3). Follow the structure of discussions in the lectures to respond to the following questions.

(a) Is the Treatment variable statistically significant in column (1)? What is the magnitude of the treatment effect? According to you, is it practically significant? Based on solely the analysis of the result in column (1), what would be the best curation/recommendation strategy for your platform?
20 points

(b) In this part, we will build on the discussion in class about the value of past visits as an input to the algorithm with additional nuance. To interpret the treatment effect of an individual with *Y* prior visits you will need to compute: $[-0.0081] + [0.0030 \times \text{Log}(Y)]$. This expression will shed light on the performance of the algorithm relative to the alternative.

Based on the results in column (2), how would your recommended strategy change relative to if you only had access to results in column (1). Describe how you arrived at the strategy using concepts of statistical and practical significance and use the expression above to support your answer with appropriate computations. **40 points**

- (c) In this final part, we will analyze the results to link to our conversation about (diminishing) marginal returns to data in Lecture 11. To do so, we add a quadratic term to the model which can give you information about the curvature as we discussed in class related to diminishing marginal returns (Hagiu and Wright in HBR).

Focusing on column (3), how does your understanding of the impact of additional data for algorithmic recommendations become more nuanced relative to the results in column (2)? To arrive at this answer, try to understand the sign, magnitude and interpretation of the coefficient on the quadratic term along with the other coefficients. **40 points**